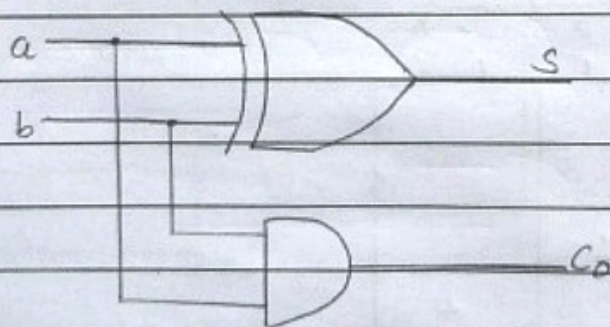
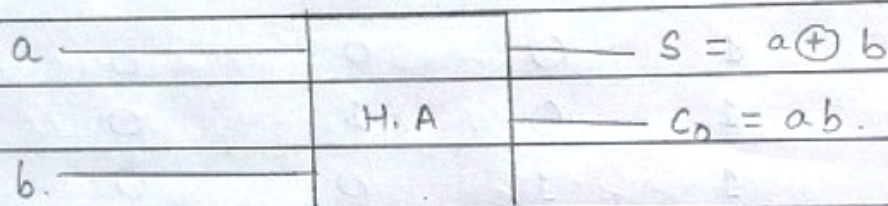
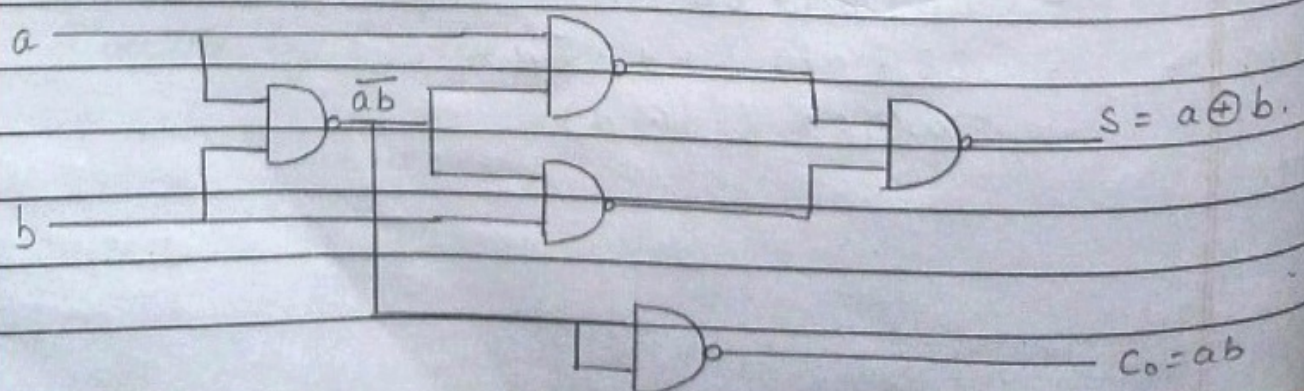


- * Other names for XNOR gate
 - coincidence circuit
 - equivalent detection
 - inclusive OR gate

	a	b	S		a	b	S	C ₀
* Half Adder.					0	0	0	0
$S = \sum m(1,2) = a \oplus b$					0	1	1	0
$C_0 = a \cdot b = \sum m(3)$					1	0	1	0
					1	1	0	1



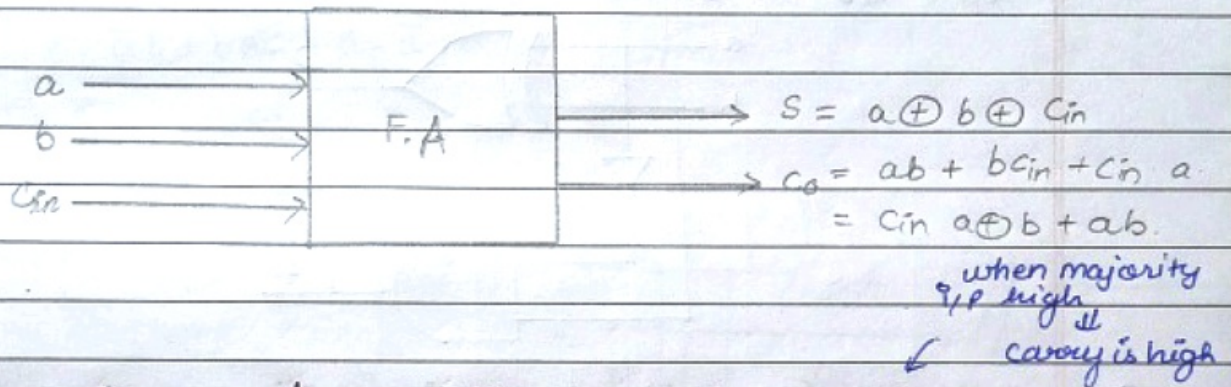
Q. Implement H.A using min no. of NAND gates.



Half adder adds only 2 bits.
It doesn't add carry.

* Full adder

H.A adds only 2 bits and hence there's no provision to add carry bit and this problem is avoided by full adder.



a	b	c_{in}	S	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

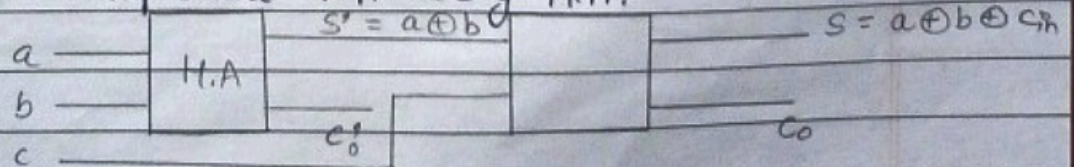
$$S = \sum m(1, 2, 4, 7) = a \oplus b$$

$$C_o = \sum m(3, 5, 6, 7) = ab + aC_{in} + bC_{in}$$

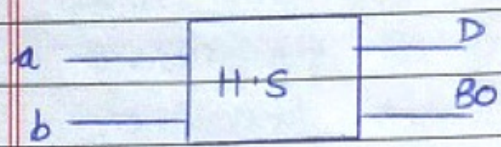
BCin

00	01	11	10
0	1	1	0
1	1	0	1

Implement F.A using H.A.

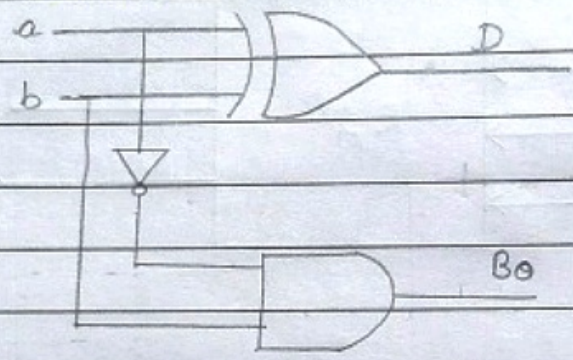


* Half Subtractor

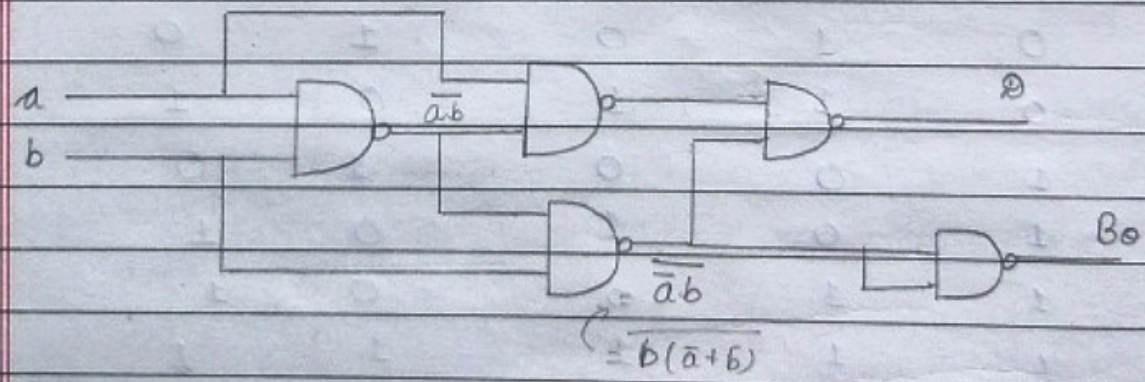


a	b	D	Bo
0	0	0	0
0	1	1	1 (add 1 to a)
1	1	0	0
1	0	1	0

$D = a \oplus b \quad \Sigma m(1, 3)$
 $Bo = \bar{a}b \quad \Sigma m(1)$

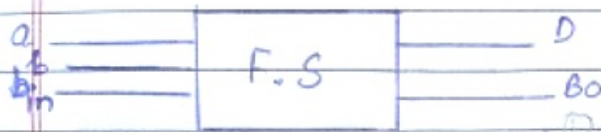


Q. Implement half subtractor using minimum no. of NAND gates.



BoC Replace a by $\bar{a}$ in F.A.) $a - (b + Bin)$

* Full Subtractor.

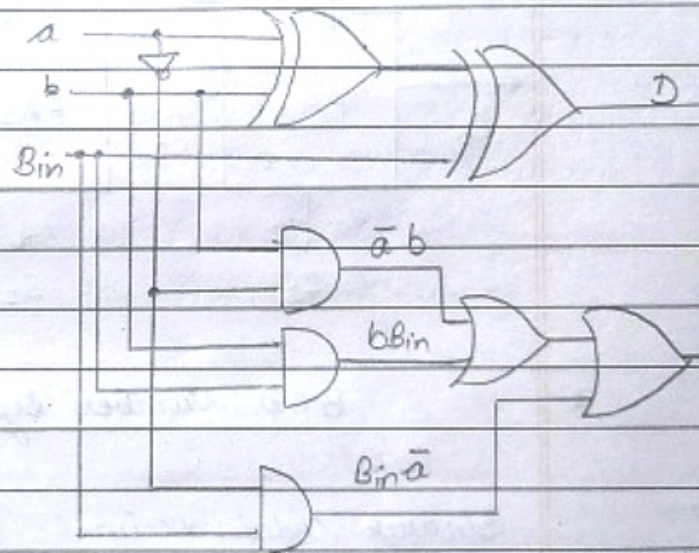


a	b	Bin	D	B ₀
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$D = \sum m(1, 2, 4, 7) \quad a \oplus b \oplus Bin$$

$$B_0 = \sum m(1, 2, 3, 7)$$

$$= \bar{a}b + bBin + Bin\bar{a}$$

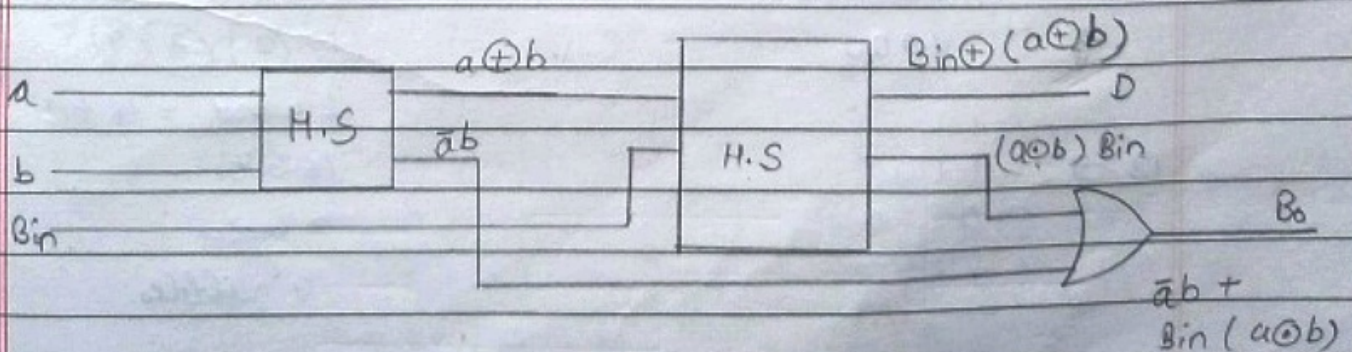


Q. Implement full subtractor using half subtractor and suitable logic gates.

$$B_0 = \sum m(1, 2, 3, 4) = \bar{a}\bar{b}Bin + \bar{a}b\bar{Bin} + \bar{a}bBin + a\bar{b}Bin$$

$$= Bin(\bar{a}\bar{b} + ab) + \bar{a}b(Bin + \bar{Bin})$$

$$= Bin(a \odot b) + \bar{a}b$$



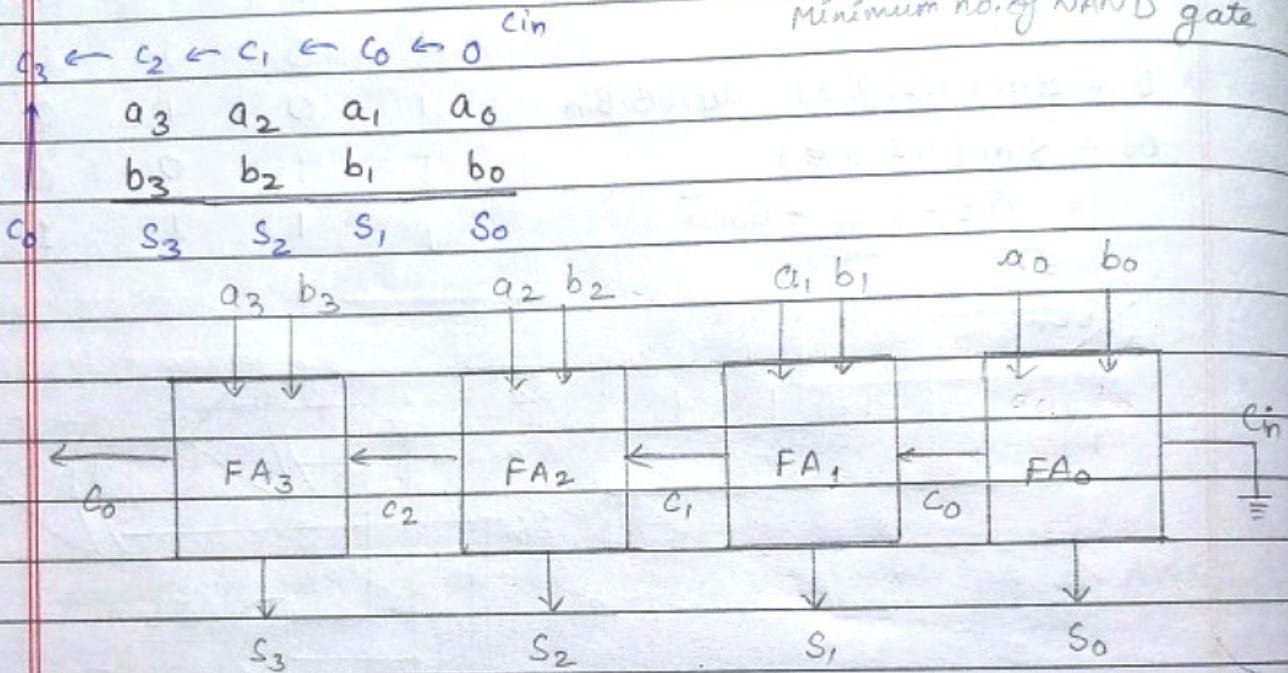
$$\begin{array}{r} 101 \\ + 0111 \\ \hline 10111 \end{array}$$

* 'n' bit parallel adder

H-Adder $\rightarrow 5$ NAND
F.A $\rightarrow 9$ NAND

To implement n bit parallel adder, we require
n full-adders or (n-1) full adder + 1 half adder.

Minimum no. of NAND gate



* BCD Number System

Binary coded decimal (BCD) has 10 digits from 0 to 9. and replace every digit with equivalent 8421 code (4 bit code)

Example:

5 $\rightarrow$ 0101

13 $\rightarrow$ 00010011

8 $\rightarrow$ 1000

~~$$\begin{array}{r} 13 > 9 \\ + 6 \\ \hline 19 \\ \hline 10011 \end{array}$$

1 3~~

(1) (2) (3) (9)

1 word = 16 bit

16 bits

2 bytes

4 nibbles

*

XS-3 code

Add 3 in every BCD digit and replace by equivalent 8421 code

Q. Determine XS-3 code of i) 5 iii) 8 ii) 15
[By default $\rightarrow$ base 10] iv) $(100011)_2$ v) $(37)_8$

i) $5 \Rightarrow 5+3 = 1000$

5 & 8 are valid BCD.

ii) $8+3 = 11 = 1011$

iii) $15 \Rightarrow 1+3 \quad 5+3 = 48 = 01001000$

iv) $(100011)_2 = (35)_{10} = 3+3 \quad 5+3 = 68$
 $= 01101000$

v) $(37)_8 = (31)_{10} = 3+3 \quad 1+3 = 64$
 $= 01100100$

	Valid	Invalid (Don't care)
BCD	0-9	10-16
XS-3	$(0+3) 3-12 (9+3)$	0, 1, 2, 13, 14, 15

* BCD Addition

After adding 2 BCD digits, if sum > 9 , add 6.
otherwise add 0.

Example:

$$\begin{array}{r} 3 \\ +5 \\ \hline \end{array} \quad \begin{array}{r} 0011 \\ 0101 \\ \hline 1000 \end{array} = 8 < 9 \quad \times \quad (8)$$

$$\begin{array}{r} 6 \\ +7 \\ \hline \end{array} \quad \begin{array}{r} 0110 \\ +0111 \\ \hline 1101 \end{array} = 13 > 9$$

$$\begin{array}{r} +0110 \quad +6 \\ \hline 10011 \end{array}$$

BCD Adder (Decimal Adder)

Q why add 6?

Max digit in BCD $\rightarrow 9$

Max digit in 4 bit $\rightarrow 15$

6

$$\begin{array}{r}
 \begin{array}{cccc}
 & A_3 & A_2 & A_1 & A_0 \\
 8 & 1 & 0 & 0 & 0 \\
 + 9 & 1 & 0 & 0 & 1 \\
 \hline
 & 1 & 0 & 0 & 0 & 1 & 17 > 9 \\
 + & & 0 & 1 & 1 & 0 & +6 \\
 \hline
 & 1 & 0 & 1 & 1 & 0 & = 8 + 6 = 14 \\
 \hline
 & 1 & & & & & 7
 \end{array}
 \end{array}$$

We require 2 parallel adder to solve BCD addition

If result > 9 , add 6.
 $10 \rightarrow 15$.

else, add 0

Correction Factor.

$S_3 S_2$ \ $S_1 S_0$	00	01	11	10
00				
01				
11	1	1	1	1
10			1	1

$$\begin{array}{cccc}
 a_3 & a_2 & a_1 & a_0 \\
 0 & 1 & 1 & 0
 \end{array}$$

$$\begin{array}{cccc}
 a_3 & a_2 & a_1 & a_0 \\
 0 & 0 & 0 & 0
 \end{array}$$

correction factor

a_1 & a_2 are same (either 0 or 1)

$$C.F = S_3 S_2 + S_3 S_1$$

NOTE:

$A_3 \ A_2 \ A_1 \ A_0 \quad B_3 \ B_2 \ B_1 \ B_0$

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